

Education

Engineering Physics – Electrical and Computer Specialization: GPA – 3.7/4.0
The University of British Columbia
Relevant Courses: Machine Learning, Stochastic Calculus and Probability, Data Structures and Algorithms, Signals and Systems, Analog/Digital Circuit Design.

Vancouver, British Columbia
Expected Graduation: May 2027

Technical Experience

Undergraduate Silicon Quantum Photonics Researcher
Stewart Blusson Quantum Matter Institute

Vancouver, British Columbia
January 2026 — Present

- Investigating the usage of nanowire single photon detectors on **silicon** for spin detection in photon-based quantum computers.
- Engineered **RF** (~3GHz) PCBAs to interface between cryogenic (4 Kelvin) photon detectors and room temperature acquisition systems, ensuring **impedance matching** and high signal to noise ratio, resulting in **high throughput** and **minimizing jitter**.
- Developed a variable low-noise current source for superconducting biasing applications, delivering up to 8mA with less than 5mA of RMS noise, resulting in stable superconductivity for cryogenic applications.
- Designed a **low-noise amplifier** to operate at 4 Kelvin with 50 MHz–3 GHz bandwidth and less than 100 mW power dissipation, enabling reliable operation in cryogenic environments while minimizing flicker and Johnson–Nyquist noise.
- Engineering a high-power **RF amplifier**, operating at 300 Kelvin, to maximise the signal to noise ratio from single photon detectors.

Electrical Engineer
UBC Thunderbots

Vancouver, British Columbia
September 2024 — Present

- Led the design and development of a **power distribution board**, isolating low (3.3 - 24V) and high (~240V) voltages, using switching, linear and flyback regulators which enhanced resistance to electrical faults and improved overall robot **safety**.
- Led the implementation of a **CAN communication interface** which increased RasPI – ESP32 communication speed by over 10x compared to the previous SPI communication, for motor control.
- Designed and validated a custom motor-driver PCB with **TIA-based phase current sensing**, enabling closed-loop **three-phase FOC** control that improved motor efficiency and reduced power consumption by **10%**.
- Designed a three-phase motor inverter, integrating **power electronics**, gate-drive circuitry, and control-oriented PCB layout.
- Developed **firmware** to interface a RaspberryPI with multiple **SoCs**, ensuring deterministic behavior through synchronized I/O interrupt handling, real-time constraints and robust handling, using **UART**, **SWD**, and **SPI**.

Undergraduate 2D Physics Researcher
University of Calgary

Calgary, Alberta
January 2025 — May 2025

- Developed and implemented a **PID-controlled** fluid cell to regulate humidity from (0%-95% relative humidity) in an atomic force microscopy setting, enabling researchers to examine the effect of humidity on 2D Materials.
- Designed and developed a full-cycle Java application using **OpenCV**, enabling visualization of wear resulting from repetitive wear.

Projects | [Portfolio](#)

Autonomous Pet Rescuing Robot — Artificial Intelligence, PCB Design, Rapid Prototyping, UART, Raspberry Pi, FreeRTOS, C++, OOP.

- Developed and designed an **H-Bridge PCB to drive two motors**, which were controlled in a **PID** algorithm with digital sensor input from a **photoelectronic circuit**.
- Spearheaded the development of an ESP-32 / Raspberry Pi GPIO board to autonomously control Stepper Motors and Line following while collecting plushies throughout a competition course.
- Trained a **YoloV11** computer vision model capable of running on a Raspberry Pi capable of recognizing plushies in diverse environments with a confidence interval of over 90%, as well as estimating central angles with <2-degree variance.
- Implemented **UART** communication between an ESP-32 and Raspberry Pi with **FreeRTOS** allowing for CV-Hardware integration.
- Utilized FreeRTOS task management to implement ESP-32 dual-core line following, allowing sampling and control intervals of < 1ms.
- Wrote firmware to control stepper motors motion with an S-Curve velocity profile resulting in the elimination of skipped steps.

Virtual Autonomous Driving — Deep Learning, TensorFlow, PyTorch, Reinforcement Learning, Computer Vision, CNN, Linux, ROS.

- Using **TensorFlow**, developed a robust convolutional neural network capable of reading words on road signs.
- Trained a **deep Q-learning** neural network model in **PyTorch** to teach a robot to autonomously navigate through a course using imitation learning, while avoiding pedestrian and vehicular collisions, with only **computer vision (OpenCV)** and lidar.
- Researched and developed a multimodal CNN-MLP architecture with late fusion, combining camera-based convolutional features and LiDAR embeddings through an **MLP fusion network** for control prediction, leading to a top score in a competition.

Skills

Engineering Software: Altium, KiCAD, GIT, LTSpice, MATLAB, Cadence Virtuoso/Allegro, MS Office, AutoCAD, MATLAB, Unix.

Programming: Python, Java, C, C++, RTL, Object Oriented Programing, OpenCV, System-Verilog, VHDL, TensorFlow, PyTorch, ZephyrRTOS.

Other Technical: Soldering, Computer Architecture, Robotics, Electronics, Troubleshooting, Circuit Design and Analysis, Linux, Robotics.